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Hawaii Seafood Guide

"Helping consumers make informed and sustainable seafood choices in Hawaii"

Introduction

Hawaii’s ocean is central to the islands’ culture, economy, and food security. Seafood, especially local fish such as Ahi, Mahimahi, and Opakapaka, plays a vital role in local cuisine. However, Hawaii’s marine ecosystems face increasing pressure from overfishing, climate change, habitat degradation, and pollution. Many residents and tourists lack clear, accessible information about which Hawaiian fish species are sustainably harvested versus those that are overfished or carry health concerns such as elevated mercury levels.

This project aims to address this information gap by creating a user-friendly web application that assists consumers to make informed and environmentally responsible seafood choices. Inspired by programs like the Monterey Bay Aquarium's Seafood Watch, the platform provides guidance tailored specifically to Hawaii's unique marine environment.

The website revolves around three core models:

Fish Species: Detailed profiles of 20+ Hawaii fish, including sustainability ratings, health notes, and preparation methods.

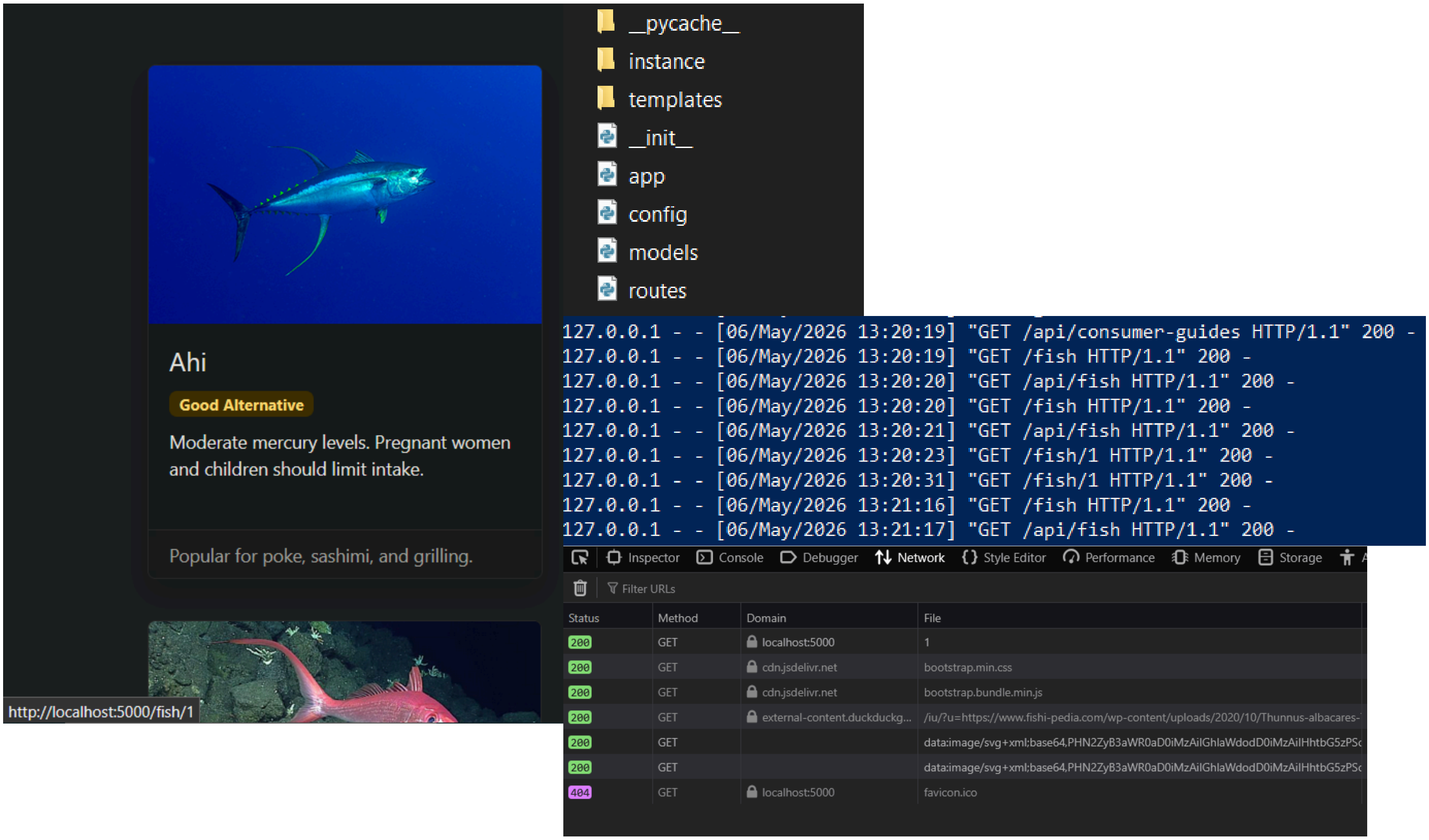
Seafood Basics: Educational content covering fishing methods, overfished areas, imported species, and the broader state of Hawaii's seafood industry.

Consumer Guides: Curated recommendations and practical guides (e.g., "Best Fish for Poke" or "Low-Mercury Options").

By combining a clean, responsive front-end built with an API developed with Flask, the application delivers dynamic information in an engaging format. The goal is not only to inform users but also to promote long-term ocean conservation and support sustainable practices that protect Hawaii's reefs and pelagic zones for future generations.

This report documents the development process, technical decisions, challenges encountered, and lessons learned throughout the creation of the Hawaii Seafood Guide.

The API



The backend of the Sustainable Hawaii Seafood Guide was built using the Flask web framework in Python. The API serves as the core layer of the application, providing structured responses that the frontend consumes to dynamically display information to users.

The API currently shows the following main endpoints:

GET / – Returns a welcome message and overview of available endpoints.

GET /api/fish – Retrieves the full list of Hawaii fish species. Supports optional query parameters for searching by name (?search=ahi) and filtering by sustainability rating (?rating=Best Choice).

GET /api/fish/<id> – Returns detailed information for a specific fish species by its unique ID.

GET /api/fish/<id>/health – Returns only the health-related information (mercury levels, consumption advice, etc.) for a given fish.

All responses are returned in clean JSON format, making the data easy to consume by the frontend. The API includes proper error handling, returning 404 responses when a requested resource is not found.

The resources are represented by URLs (e.g., /fish/1), and HTTP methods define the action (primarily GET for read operations).

In its current implementation, fish species data is stored in-memory within the application. Future improvements will include integrating a database layer using Flask-SQLAlchemy to allow persistent storage, easier updates, and support for further operations.

The API is fully containerized using Docker, ensuring consistent deployment across different environments. This approach supports the design principles of portability and generalization, allowing the backend to be developed, tested, and deployed independently of the frontend.

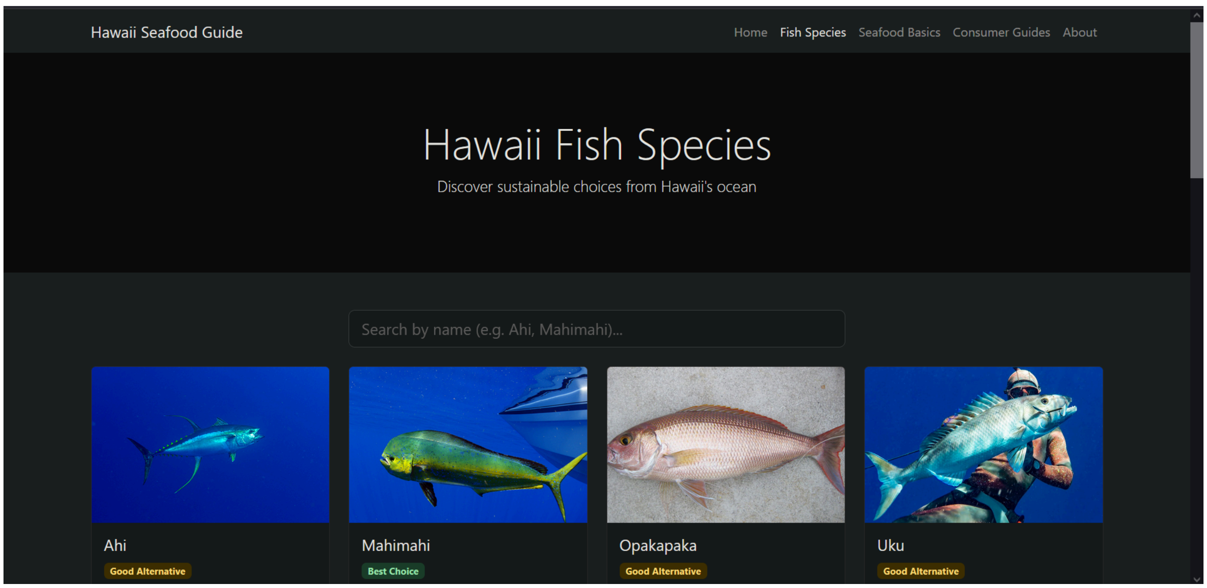
The Front-end Design

The front-end of the Hawaii Seafood Guide was designed with simplicity, minimalism, and educational value in mind. We used **HTML, CSS, and JavaScript** with the **Bootstrap** framework to create a clean, modern, and fully responsive user interface that works well on both desktop and mobile devices.

Key Pages

1. Homepage / Landing Page

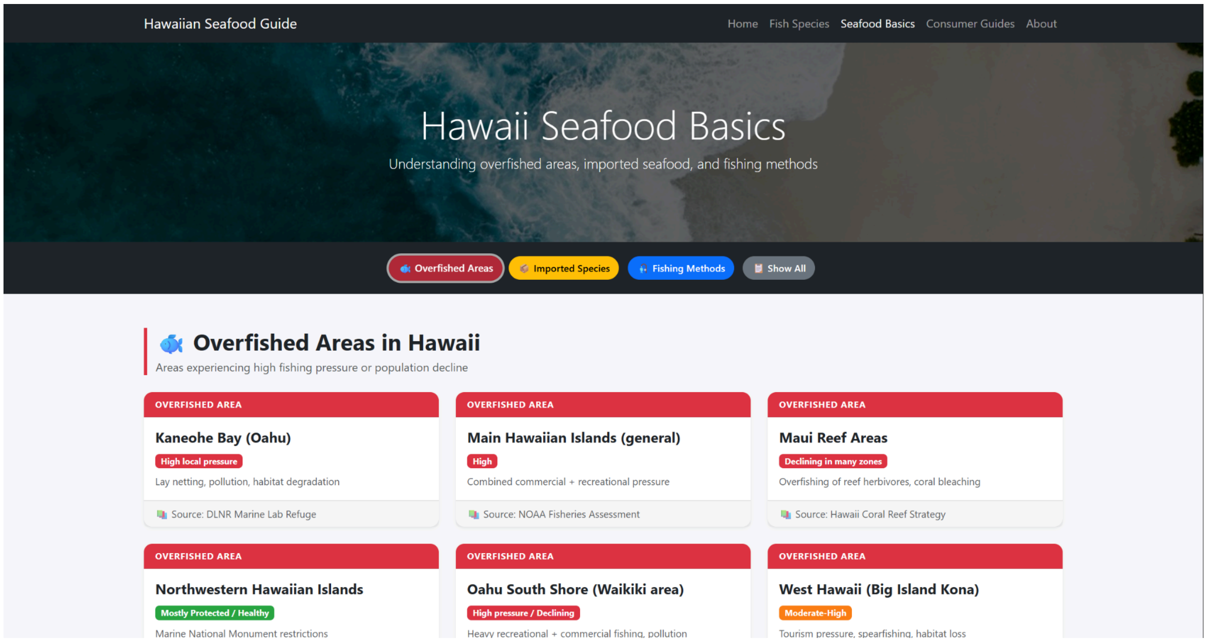
- Features a large hero section with a beautiful ocean background and a clear explanation on what each page is about.
- Includes three prominent buttons that link directly to the three main models (Fish Species, Seafood Basics, and Consumer Guides).
- Simple and welcoming layout designed to immediately communicate the purpose of the website.



2. Fish Species Page

- The main feature of the front-end.
- Displays all 20 fish species in a responsive grid of Bootstrap cards.
- Each card includes: fish image, common and Hawaiian name, sustainability rating (color-coded badges), health notes, and preparation tips.
- Includes a live search bar at the top that filters fish in real-time as the user types.
- Hover effects and clean typography make the page engaging and easy to browse.

3. Seafood Basics Page



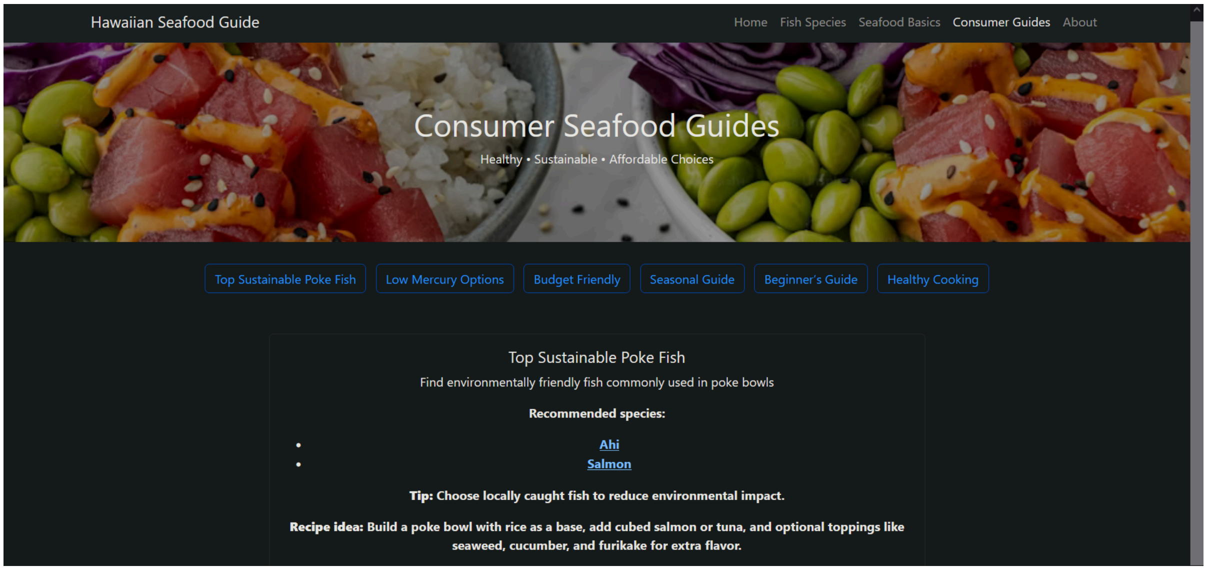
- Provides general educational content about Hawaii's seafood industry, sustainability challenges, and key background information.

Features three main sections:

- **Fishing Methods**: Describes common techniques used in Hawaii (e.g., spearfishing, pole-and-line, longlining, lay netting) with sustainability rankings and environmental impact explanations.
- **Imported Species**: Highlights the most commonly imported seafood to Hawaii (shrimp, Atlantic salmon, tilapia, etc.) and discusses why they are imported and their sustainability considerations.
- **Overfished Areas**: Details specific regions in Hawaii facing heavy fishing pressure (e.g., Waikiki, Kona, Kaneohe Bay) and the main causes (recreational fishing, commercial pressure, habitat loss, and pollution).

The page uses clear tables and to make the information easy to understand for general users. This aims to give users broader context beyond individual fish species, helping them understand the bigger picture of Hawaii's marine ecosystem.

4. Consumer Guides Page



This page offers practical, curated recommendations to help users make better seafood choices in everyday situations.
Contains multiple guides such as:

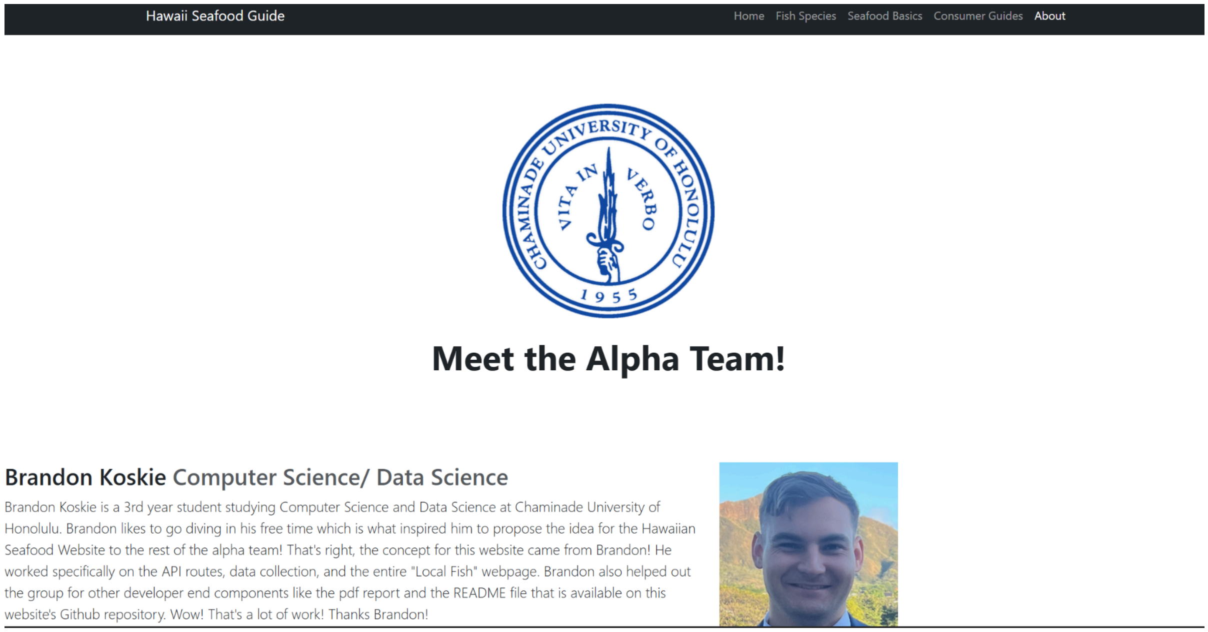
- "Top Sustainable Poke Fish"
- "Low-Mercury Options"
- "Budget Friendly"

Each guide includes:

- Recommended species
- Preparation tips and recipes
- Some environmental tips

The guides focus on translating sustainability data into actionable advice for local residents, tourists, and home cooks.

5. About Page



This page offers insight into the person behind the developing team. Each Alpha Team member has a short bio and headshot along with their contact information for other developers who are interested in collaborating in the future. The bios themselves give brief insights into the team's personal hobbies and professional careers.

Technologies Used

- **Bootstrap 5**: For responsive grid system, cards, navbar, and styling.
- **JavaScript (Fetch API)**: To dynamically load fish data from the backend API.
- **Custom CSS**: For additional styling, hover effects, and color-coded sustainability ratings.

The front-end successfully integrates with the Flask API, demonstrating a clean relationship between backend data logic and frontend presentation.

The Database

The Hawaii Seafood Guide uses a structured data model design to organize and serve information. We implemented a hybrid approach combining in-memory data structures with partial integration of a database layer using Flask-SQLAlchemy.

Implementations

- **Fish Species Model**: Currently stored as a list of dictionaries directly in routes.py. This in-memory approach allowed for rapid development and testing of the frontend fish cards page. The model includes key attributes such as id, common_name, scientific_name, sustainability_rating, health_notes, preparation, description, and image_url. Twenty species have been manually entered with sourced information.

- **Seafood Basics Model**: Data was initially collected and organized in Google Sheets for easy collaboration. It has been partially converted into a format compatible with SQLAlchemy. This model covers fishing methods, overfished areas, imported species, and general educational content about Hawaii's seafood industry.

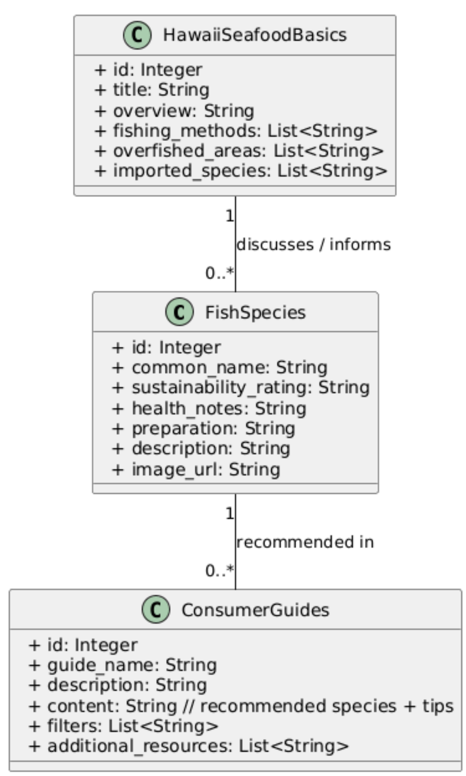
- **ConsumerGuides Model**: Similar to the Fish Species model, this data is currently stored as in-memory Python structures in the routes file. It includes guide titles, descriptions, and recommended species.

The models are designed with clear relationships:

- Fish Species is the central model.
- Consumer Guides references multiple Fish Species entries.
- Hawaii Seafood Basics provides contextual information that supports and informs both of the other models.

This hybrid approach (in-memory + partial SQLAlchemy) allowed us to meet project deadlines while building toward a more robust, production-ready database implementation.

- **UML Diagram:**



This database design supports the REST API architecture and allows the frontend to load content.

Challenges, What Worked Well, and Lessons Learned

Challenges:

- Coordinating team work
- Learning Flask routing and JSON handling
- Finding good quality images and reliable sustainability data

What worked well:

- Flask + Bootstrap integration
- Dynamic frontend with JavaScript fetch
- Clear API design

Lessons learned:

- Importance of clean project structure
- How to create and customize an HTML file
- using virtual environments, etc.
- How to create a website

Work Details (Team Contributions)

- Brandon: API routes, Fish Species model, Seafood Basics data, creation of website structure/folders, PDF report, Fish Species frontend design, presentation slides.

- Tallen: Home Page and About Page frontend and backend design, API routes, presentation slides.

- DJ: Seafood Basics frontend and backend design, API routes, API test points.

- Kamryn: Consumer Guide frontend and backend design, API routes.

Future Improvements

- Update the database so that it primarily uses Flask-SQLAlchemy
- Admin panel to add new fish/guides
- Add more species to the list

Conclusion

The Hawaii Seafood Guide project successfully demonstrates the integration of web development technologies to address a real-world environmental and cultural issue in Hawaii. Over the course of this project, our team built a functional web application featuring a Flask API, Bootstrap frontend, and structured data models focused on Hawaii fish species, seafood basics, and consumer guidance.

We achieved our primary goal which was creating an accessible, informative platform that helps users make more sustainable seafood choices. The application currently provides detailed profiles for 20 Hawaiian fish species, educational content on fishing methods and overfished areas, and foundational consumer guides that are all delivered through a clean, responsive user interface.

This project allowed us to apply and strengthen many of the skills learned throughout the semester, including API design, Flask development, frontend integration with JavaScript, error handling, containerization, and collaborative software development using GitHub. The structure we implemented makes the application easy to maintain and improve upon in the future.

While the current version represents a solid minimum viable product, there remains significant potential for growth. Future iterations could include a full database implementation with Flask-SQLAlchemy and expanded educational content.

Ultimately, this project reinforced the importance of combining technology with environmental awareness. By making sustainability information clear and engaging, we hope this platform can contribute, even in a small way, to the protection of Hawaii's marine resources and the continuation of its traditions for generations to come.